Schema files are also useful if you want a quick look at what attributes an Active Record object has. This information is not in the model's code and is frequently spread across several migrations, but the information is nicely summed up in the schema file. The annotate_models gem automatically adds and updates comments at the top of each model summarizing the schema if you desire that functionality.

6.2 Types of Schema Dumps

There are two ways to dump the schema. This is set in `config/application.rb` by the `config.active_record.schema_format` setting, which may be either `:sql` or `:ruby`.

If `:ruby` is selected then the schema is stored in `db/schema.rb`. If you look at this file you'll find that it looks an awful lot like one very big migration:

```
ActiveRecord::Schema.define(version: 20080906171750) do
  create_table "authors", force: true do |t|
    t.string   "name"
    t.datetime "created_at"
    t.datetime "updated_at"
  end

  create_table "products", force: true do |t|
    t.string   "name"
    t.text "description"
    t.datetime "created_at"
    t.datetime "updated_at"
    t.string "part_number"
  end
end
```

In many ways this is exactly what it is. This file is created by inspecting the database and expressing its structure using `create_table`, `add_index`, and so on. Because this is database-independent, it could be loaded into any database that Active Record supports. This could be very useful if you were to distribute an application that is able to run against multiple databases.

There is however a trade-off: `db/schema.rb` cannot express database specific items such as triggers, stored procedures or check constraints. While in a migration you can execute custom SQL statements, the schema dumper cannot reconstitute those statements from the database. If you are using features like this, then you should set the schema format to `:sql`.

Instead of using Active Record's schema dumper, the database's structure will be dumped using a tool specific to the database (via the `db:structure:dump` Rake task) into `db/structure.sql`. For example, for PostgreSQL, the `pg_dump` utility is used. For MySQL, this file will contain the output of `SHOW CREATE TABLE` for the various tables.

Loading these schemas is simply a question of executing the SQL statements they contain. By definition, this will create a perfect copy of the database's structure. Using the `:sql` schema format will, however, prevent loading the schema into a RDBMS other than the one used to create it.

6.3 Schema Dumps and Source Control

Because schema dumps are the authoritative source for your database schema, it is strongly recommended that you check them into source control.

`db/schema.rb` contains the current version number of the database. This ensures conflicts are going to happen in the case of a merge where both branches touched the schema. When that happens, solve conflicts manually, keeping the highest version number of the two.

7 Active Record and Referential Integrity